

Lean Agent Code & Multi-Agent Systems

Ponytail · gitea-mcp · Claude Code · Self-Hosted LLMs

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Modular Quantum Solutions

Lean Agent Code

Why Code Quality Matters for AI Agents

The problem: AI agents tend to over-engineer — adding layers, abstractions, and boilerplate that create maintenance debt without adding value.

Research findings on AGENTS.md / code-guidance files:

Intervention	Compliance
Documentation alone	~25–40%
Runtime enforcement hooks	~95%
Auto-generated AGENTS.md	<i>reduces success rate by ~3%</i>

Conclusion

Rules without enforcement drift. The solution is a decision ladder baked into every coding session.

Ponytail Decision Ladder

Before writing a single line of code, an AI agent asks:

1. **Does it need to exist?** — skip if not required
2. **Already in codebase?** — reuse existing code
3. **Stdlib handles it?** — prefer standard library
4. **Native platform feature?** — use platform primitives
5. **Installed dependency?** — use what's already there
6. **One line?** — write one line
7. **Then:** minimum viable implementation

Non-negotiable

Validation, error handling, security, and accessibility are always kept — ponytail targets complexity, not safety.

Ponytail in Practice — QPUBench Examples

Fixes made during QPUBench development under ponytail review:

File	Issue	Fix
store.py	pd.Series mask — bug + unnecessary	Replaced with <code>df = df[df[col] == val]</code> loop
stub.py	pattern = None — dead variable	Removed
StubMBQCAadapter.spec adapter.py (qforte)	<code>tuple([sym, tuple(xyz)])</code> — list wrapped in tuple	Changed to <code>(sym, tuple(xyz))</code>
store.py _iter	No return type	Added <code>-> Iterator[BenchmarkRecord]</code>
record.py _utcnow	Lambda assigned to name (noqa E731)	Converted to typed def

Python Quality Toolchain — QPUBench

```
ruff check src/ tests/      # lint (line-length 100, target py311)
ruff format src/ tests/    # format
mypy src/                  # static types (strict = true)
pytest tests/ -q           # 259 tests, SDK-dependent ones skip cleanly
```

Tool	Role
Pydantic v2	Runtime-validated schemas — field types enforced at parse time
ruff	Linting + auto-formatting, replaces flake8/isort/black
mypy strict	Full type coverage — Any only where unavoidable
pytest	Schema-only test suite runs with <code>pip install .</code> alone

Ponytail — Install & Use

```
# Claude Code plugin
/plugin install ponytail@ponytail

# Review current diff for over-engineering
/ponytail-review

# Scan entire repo for unnecessary code
/ponytail-audit

# Collect deferred optimisation shortcuts
/ponytail-debt
```

Supported agents: Claude Code · OpenAI Codex · GitHub Copilot CLI · Cursor · Windsurf · Gemini CLI · Devin

Multi-Agent System

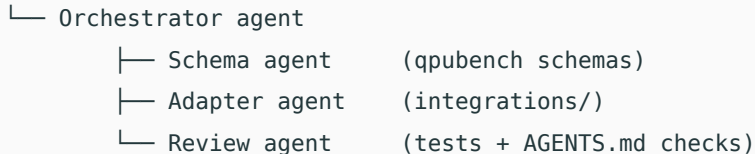
Why Multi-Agent?

Single-agent limitation: one agent, one context window, one repo at a time.

Multi-agent enables:

- Parallel work across multiple repos or services
- Specialised agents per domain (schemas · adapters · tests · docs)
- Continuous background tasks (CI watching, PR review, dependency updates)
- Self-hosted pipelines with no cloud data exposure

User



gitea-mcp Integration

gitea-mcp connects AI agents directly to Gitea repositories via MCP

AI Agent (Claude Code / Cursor / Windsurf)

└─ MCP protocol

└─ gitea-mcp server (gitea.com/gitea/gitea-mcp)

└─ Gitea REST API

└─ repos · issues · PRs · wikis · CI

- Official MCP server from the Gitea project
- Agents can read/write code, open PRs, comment on issues, trigger CI
- Works with Claude Code, Mistral Vibe, Gemini CLI, OpenCode, Devin
- Self-hosted Gitea = full data sovereignty

Supported Agents & Platforms

Agent	Open-weight	Self-hostable	gitea-mcp
Claude Code (Anthropic)	No	No	Yes
Mistral Vibe	Yes	Yes	Yes
Gemini CLI (Google)	No	No	Yes
GitHub Copilot CLI	No	No	Yes
OpenCode	Depends	Yes	Yes
Devin	No	No	Yes

All agents read the same `AGENTS.md` file — one set of rules, any tool.

Claude Code + Mistral Vibe

Claude Code (*Anthropic*)

- CLI-first agentic coding assistant
- CLAUDE.md / AGENTS.md for persistent rules
- Git worktrees for isolated parallel agents
- Per-project memory system across sessions
- Hooks for automated validation

Mistral Vibe (*Mistral AI*)

- Open-weight models, self-hostable
- MCP-compatible via gitea-mcp
- Suitable for offline / air-gapped environments
- No data leaves your infrastructure

Both integrate with the same AGENTS.md instruction file.

Context & Memory Management

Per-session context discipline:

File	Purpose
AGENTS.md	Language-agnostic, permanent agent instructions
CLAUDE.md	Claude-specific rules, project conventions
CONTEXT.md	Current task, status, constraints — written before each session

Workflow:

- Before starting → write CONTEXT.md with current task + constraints
- Each session → "Read CONTEXT.md and CLAUDE.md first"
- When switching repos → update CONTEXT.md with what was just completed
- Parallel agents → clear boundary definitions + shared contract.ts

Multi-Agent Memory Architecture

Four persistent memory types (survive across sessions):

Type	Content	When to save
user	Role, expertise, preferences	First conversation
feedback	Corrections + validated approaches	After any feedback
project	Goals, deadlines, decisions	When context changes
reference	External system pointers	When a resource is named

```
~/.claude/projects/<repo>/memory/
```

```
├─ MEMORY.md          <- index (always loaded, max 200 lines)
```

```
├─ user_role.md
```

```
├─ feedback_testing.md
```

```
└─ project_schema.md
```

Self-Deployed LLMs

Abzu (HuggingFace) — open-weight models for quantum computing tasks

- Local deployment via HuggingFace transformers / Ollama / vLLM
- No data leaves your infrastructure
- Suitable for air-gapped laboratory environments
- Combine with self-hosted Gitea + gitea-mcp for a fully sovereign pipeline

Local LLM (Ollama / vLLM)

└─ MCP client (Claude Code / Cursor)

└─ gitea-mcp --> Gitea (self-hosted)

└─ qpubench repo

Full self-hosted stack

Model · version control · CI · agent memory — nothing leaves your network.

Summary

Key Takeaways

Lean code (ponytail)

- Decision ladder before every change
- Runtime enforcement » documentation
- Remove dead code ruthlessly
- Type hints everywhere
- 46% fewer lines, 22% fewer tokens

Links: github.com/DietrichGebert/ponytail · gitea.com/gitea/gitea-mcp · huggingface.co/Abzu

Multi-agent systems

- gitea-mcp bridges agents to Gitea
- Same AGENTS.md for all tools
- CONTEXT.md per session
- Four memory types persist state
- Full self-hosted stack possible